

Single-Gene NIPT: Fetal Variant Detection and In Silico Sample Simulation

Twist UCL TE-96276661 GRCh38 183 genes · 5,138 regions

Abstract

Single-Gene Non-Invasive Prenatal Testing (sgNIPT) identifies fetal pathogenic variants from cell-free DNA (cfDNA) in maternal plasma. Because fetal DNA represents only a small fraction of the total cfDNA pool (the fetal fraction, FF), variant allele frequencies (VAF) of fetal-specific variants are predictably low — approximately $FF/2$. This white paper describes the sgNIPT pipeline's fetal fraction estimation algorithm, variant origin classification logic, and the in silico simulation framework used to generate validation samples **FF10_sgNIPT_hotspot** and **FF10_sgNIPT_dominant_xlinked**.

1. Biological Model of cfDNA in NIPT

Maternal plasma cfDNA is a mixture of maternal-derived and placenta-derived (fetal) DNA fragments. At any genomic locus, the observed VAF reflects both contributions:

Observed VAF = (maternal contribution × maternal AF) + (FF × fetal AF)

For a locus where the mother is homozygous reference (hom-ref) and the fetus carries a heterozygous pathogenic allele (the canonical "paternal inheritance" scenario), the fetal allele is present in $FF/2$ of all cfDNA molecules:

Observed VAF $\approx$ FF / 2 (e.g. FF = 10% $\rightarrow$ VAF $\approx$ 5%)

For maternal heterozygous variants, the allele is present at $\sim 50\%$ in maternal DNA and shifts by $\pm FF/2$ depending on fetal inheritance:

Fetus inherits maternal ALT
cfDNA VAF $\approx$ 0.5 + FF/2 (upward shift)
Fetus is REF (did not inherit)
cfDNA VAF $\approx$ 0.5 – FF/2 (downward shift)

2. Fetal Fraction Estimation

Fetal fraction is estimated from informative SNP sites identified in a panel-wide pileup, computed by fetal_fraction.py. The result is passed as --fetal-fraction to the downstream variant classifier.

2.1 Informative SNP Selection

A site is informative if the maternal genotype is homozygous-alternative (GT=1/1) and the fetus inherits one reference allele from the father. The minor allele in cfDNA (the REF allele from fetal DNA) appears at frequency $FF/2$.

Selection criteria (fetal_fraction.py DEFAULT_CONFIG):

MIN DEPTH AT SITE 50x
MINOR AF RANGE 0.005-0.25
MIN INFORMATIVE SNPS 10
MIN FF TO REPORT 2%
MAX FF TO REPORT 50%
OUTLIER REMOVAL IQR×1.5

2.2 Estimation Formula

For each informative site *i*, the per-SNP FF estimate is:

$$FF_i = 2 \times \text{minor_AF}_i \quad \text{where } \text{minor_AF} = \min(\text{alt_AF}, 1 - \text{alt_AF})$$

Primary FF = median(FF_i) after IQR-based outlier removal
95% CI: normal approximation $\pm$ 1,000-iteration bootstrap (seed 42)

2.3 Supplementary Estimation Methods

Three additional methods provide cross-validation. Discordance ≥ 0.10 triggers a warning.

METHOD	FORMULA	NOTE
FF SNPs (primary)	$2 \times \text{median}(\text{minor_AF})$	IQR outlier-filtered; bootstrap CI
Chr X ratio	$2 \times (1 - \text{chrX_ratio} / \text{female_baseline})$	Male fetus only; baseline = 1.0
Chr Y density	$2 \times \text{Y_density} / \text{autosome_density}$	Unreliable for small panels
Fragment size	$\text{short_ratio} \times 0.5$	short < 150 bp, long $\geq$ 250 bp

3. Variant Origin Classification

After variant calling with `bcftools mpileup` on panel target regions, each variant is classified by `variant_analysis.py` (`classify_variant_origin`). The two primary inputs are the observed VAF and the estimated FF.

3.1 Tolerance Window

$$\begin{aligned} \text{expected_fetal_vaf} &= FF / 2 \\ \text{tolerance} &= \text{expected_fetal_vaf} \times \text{fetal_af_tolerance_factor} \quad (\text{default } 2.0) \\ &= FF \end{aligned}$$

At FF=10%: $\text{expected_fetal_vaf} = 0.05$, $\text{tolerance} = 0.10$
Fetal-specific window: $VAF \in [0, 0.15]$ (and binomial test must pass)

3.2 Classification Decision Tree (in evaluation order)

ORIGIN	VAF RANGE	MATERNAL GT	FETAL GT	CONFIDENCE	GATE
fetal_specific	$\approx FF/2$ (~5%)	hom_ref	het	high	Binomial $p < 0.01$ vs $p_0=0.001$

maternal_het	0.35 – 0.65	het	uncertain / likely_het / likely_ref	medium	Deviation from 0.5 vs FF/2
maternal (hom_alt)	≥ 0.85	hom_alt	at_least_het	high	VAF threshold
possible_fetal	$\approx \text{FF}/2$	hom_ref	uncertain	low	VAF in window, binomial not significant
noise	< 0.01	—	—	—	Below $2 \times \text{min_vaf}$
ambiguous	other	—	—	low	Does not fit any class

3.3 Fetal-Specific Binomial Test

A VAF within the tolerance window is necessary but not sufficient for a high-confidence fetal_specific call. A one-sided binomial test is applied against a null error rate of $p_0 = 0.001$ (sequencing noise floor):

```
H0: P(alt) = 0.001      (background sequencing noise)

z = (observed_alt_AF - p0) / sqrt(p0 * (1 - p0) / depth)
p = 0.5 * erfc(z / sqrt(2))

p < 0.01  → origin = fetal_specific,   confidence = high
p ≥ 0.01  → origin = possible_fetal,   confidence = low
```

3.4 Maternal Heterozygous: Fetal Inheritance Sub-classification

For maternal_het variants (VAF 0.35-0.65), a secondary test assesses fetal inheritance by comparing the VAF deviation from 0.5 against the expected FF/2 shift:

```
deviation = VAF - 0.5

|deviation| < tolerance          → fetal GT = uncertain   (cannot determine)
deviation > 0  and |deviation - FF/2| < tolerance → fetal GT = likely_het   (fetus inherited ALT)
deviation < 0  and |deviation + FF/2| < tolerance → fetal GT = likely_ref   (fetus is REF)
```

4. Pipeline Parameters Reference

PARAMETER	DEFAULT VALUE	SOURCE
<code>min_depth</code>	10	variant_analysis.py
<code>min_vaf</code>	0.005	variant_analysis.py
<code>maternal_hom_alt_threshold</code>	0.85	variant_analysis.py
<code>maternal_het_lower</code>	0.35	variant_analysis.py
<code>maternal_het_upper</code>	0.65	variant_analysis.py
<code>fetal_af_tolerance_factor</code>	2.0	variant_analysis.py
<code>binomial_pvalue_threshold</code>	0.01	variant_analysis.py
<code>binomial_null_p</code>	0.001	variant_analysis.py
<code>ff_min_depth (FF SNPs)</code>	50	fetal_fraction.py
<code>ff_min_minor_af</code>	0.005	fetal_fraction.py
<code>ff_max_minor_af</code>	0.25	fetal_fraction.py
<code>ff_min_informative_snps</code>	10	fetal_fraction.py
<code>bootstrap_iterations</code>	1,000	fetal_fraction.py
<code>vc_min_depth</code>	50	nextflow.config
<code>vc_min_alt_count</code>	2	nextflow.config
<code>ff_spike_min_depth</code>	30	simulate_nipt_bam.py

vc_min_depth vs post-call filter depth

`vc_min_depth=50` applies at variant calling time (nextflow.config); sites below this depth are not emitted by bcftools. `min_depth=10` in `variant_analysis.py` is a post-call filter on the VCF. In practice, the 50× calling threshold is the binding constraint.

5. In Silico cfDNA BAM Simulation

`simulate_nipt_bam.py` injects synthetic reads at defined loci into a real maternal panel BAM (`.target.bam`), faithfully reproducing the low-VAF cfDNA signal of fetal-specific variants.

5.1 Spike-in Arithmetic

```
target_VAF = FF / 2
n_to_add   = max(1, round(n_existing × target_VAF / (1 - target_VAF)))
```

Example: depth = 1000, FF = 10%

```
target_VAF = 0.05
n_to_add   = round(1000 × 0.05 / 0.95) ≈ 53 synthetic reads
```

5.2 Read Cloning Algorithm

For each synthetic read to add, the algorithm:

1. Selects a random template read overlapping the target position.
2. Maps the reference position to query index via CIGAR alignment pairs (`get_aligned_pairs`).
3. Clones the `AlignedSegment` ; renames `query_name` to `{original}_sim_{i}` .
4. Replaces the base at the query position with the spike base (ALT or REF).
5. Clears the duplicate flag (`flag & ~0x400`) to prevent MarkDuplicates removal.
6. Preserves CIGAR, mapping quality, base qualities, and all auxiliary tags.

After all reads are appended, the BAM is sorted and indexed. Original reads are never modified.

5.3 Two Spike Categories

Disease / Pathogenic Variants

Source: `--variants TSV`
Spike base: **ALT**
Maternal precondition: **hom_ref** (ref_frac ≥ 0.90)
Non-hom-ref sites are **skipped**
Models: paternal het fetal allele at VAF ≈ FF/2

FF-Informative SNPs

Source: `--ff-vcf` (GT=1/1 sites)
Spike base: **REF**
Maternal precondition: **hom_alt** (ref_frac ≤ 0.10)
Min depth: 30× (`--ff-min-depth`)
Models: fetal het adding REF allele at VAF ≈ FF/2

6. Sample: FF10_sgNIPT_hotspot

FETAL FRACTION

10%

TARGET SPIKE VAF

5%

HOTSPOT SNVS

22

FF SNPS SPIKED

100

6.1 Creation Workflow

The hotspot sample was built from a real completed run, not from another simulated BAM:

- **Source BAM:** analysis/2603/Test_sgNIPT/alignment/Test_sgNIPT.target.bam
- **FF SNP VCF:** analysis/2603/Test_sgNIPT/fetal_fraction/Test_sgNIPT.ff_variants.vcf
- **Disease variants:** data/spike-in/hotspot_variants.tsv (22 SNVs)
- **Output:** data/test/sim_bam/FF10_sgNIPT_hotspot.target.bam
- **Parameters:** --fetal-fraction 0.10 --n-ff-snps 100 --seed 42

The hotspot variant list was derived from singleGene_NIPT_with_hotspots.xlsx by build_variants_tsv.py : RASopathy and oncogenic hotspot mutations were matched to panel BED coordinates on GRCh38 and REF/ALT verified via reference lookup. One HRAS site (chr11:534289) may have been skipped at spike time due to insufficient reads in the source BAM.

6.2 Spiked Hotspot Variant List

GENE	CHROM	POSITION	REF	ALT	HGVS C.
SLC25A24	chr1	108157481	C	T	c.650G>A
SLC25A24	chr1	108157482	G	A	c.649C>T
NRAS	chr1	114716127	C	T	c.35G>A
RIT1	chr1	155904739	C	T	c.269G>A
RIT1	chr1	155904798	G	C	c.170C>G
LMNA	chr1	156136984	C	T	c.1824C>T
SHOC2	chr10	110964362	A	G	c.4A>G
FGFR2	chr10	121520163	G	C	c.755C>G
HRAS	chr11	534289	C	T	c.34G>A (may be skipped)
KRAS	chr12	25245345	C	T	c.35G>A
PTPN11	chr12	112477719	A	G	c.922A>G
SOS2	chr14	50182521	A	C	c.800T>G
MAP2K1	chr15	66436843	A	G	c.389A>G
STAT3	chr17	42329643	G	A	c.1144C>T
SOS1	chr2	39007168	C	T	c.1300G>A
RAF1	chr3	12604200	G	A	c.770C>T
FGFR3	chr4	1801837	C	T	c.742C>T
FGFR3	chr4	1804372	A	G	c.1118A>G
FGFR3	chr4	1804392	G	A	c.1138G>A/C
BRAF	chr7	140801502	T	C	c.770A>G

FGFR1	chr8	38424690	G	C	c.755C>G
MECP2	chrX	154031355	G	A	c.502C>T

6.3 Portal Review Interpretation

Total analyzed: 370 · Maternal het: 211 · Maternal hom: 137 · Fetal: 6

The 211 maternal het and 137 maternal hom variants are **real germline variants already present in Test_sgNIPT.target.bam**. They were not introduced by the spike-in. The 6 fetal variants represent hotspot spike-in sites that survived variant calling (`vc_min_depth=50`) and binomial classification at ~5% VAF. Recovery of 6 out of ~21 attempted spikes is expected — not all sites achieve sufficient depth or cross the binomial significance threshold at 5% VAF.

7. Sample: FF10_sgNIPT_dominant_xlinked

FETAL FRACTION
10%

TARGET SPIKE VAF
5%

DISEASE SNVS
871

FF SNPS SPIKED
100

This sample uses the same simulation engine and source BAM as FF10_sgNIPT_hotspot, but targets 871 known pathogenic missense/nonsense variants across dominant and X-linked disease genes from `sgNIPT_dominant_xlinked_pathogenic_cmutations.xlsx`, converted to `dominant_xlinked_variants.tsv` by `build_variants_tsv.py`.

7.1 Comparison Table

	FF10_SGNIPT_HOTSPOT	FF10_SGNIPT_DOMINANT_XLINKED
Variant TSV	hotspot_variants.tsv	dominant_xlinked_variants.tsv
Spike count	22	871
Source Excel	singleGene_NIPT_with_hotspots.xlsx	sgNIPT_dominant_xlinked_pathogenic_cmutations.xlsx
Fetal variants (portal)	6	9
Total analyzed	370	373
Maternal het	211	211
Maternal hom	137	137

The identical maternal het/hom counts confirm both samples share the same source BAM. The low fetal recovery rate (hotspot: 27%, dominant_xlinked: 1%) reflects the 50x calling depth threshold and binomial significance requirement at 5% VAF.

8. Standard Test Sample Generation (`make_test_nipt_samples.sh`)

The official script builds the FF5/FF10/FF15 test suite using the same simulator with the 6-variant `data/test/test_variants.tsv` :

```
for FF in 5 10 15; do
  simulate_nipt_bam.py \
    --bam          Test_sgNIPT.target.bam \
    --output       FF${FF}_sim_Test_sgNIPT.target.bam \
    --fetal-fraction 0.${FF} \
    --variants     data/test/test_variants.tsv \
    --ff-vcf       Test_sgNIPT.ff_variants.vcf \
    --n-ff-snps   100 --seed 42
done
```

The hotspot and dominant_xlinked samples were created using the same pattern with custom variant TSVs, manually executed via Docker. Both output BAM samplesheet CSVs are consumed by `run_sgnipt.sh --input-bam` mode, which bypasses FASTQ-to-BAM steps.

9. Pipeline Architecture Overview

The full sgNIPT pipeline (Nextflow v1.2.0, Docker image `sgnipt:latest`) operates in two modes:

FASTQ Mode (clinical samples)
1. fastp QC (Q15, min 36 bp) 2. BWA-MEM2 alignment → GRCh38 3. Picard MarkDuplicates 4. samtools target extraction 5. Fetal fraction estimation 6. bcftools variant calling 7. Variant origin classification 8. VEP annotation 9. Report generation
BAM Mode (simulation / re-analysis)
1. — (fastp skipped) 2. — (BWA skipped) 3. — (MarkDup skipped) 4. — (extraction skipped) 5. Fetal fraction estimation 6. bcftools variant calling 7. Variant origin classification 8. VEP annotation 9. Report generation

MarkDuplicates and simulated BAMs

Simulated BAMs must be processed in BAM mode (which skips MarkDuplicates) because synthetic reads share the same coordinates as their template reads. Running MarkDuplicates on a spiked BAM would remove cloned reads as optical/PCR duplicates, destroying the intended spike-in signal. This is why `simulate_nipt_bam.py` uses `.target.bam` (post-markdup) as input, and the samplesheet triggers BAM-mode entry in the pipeline.

10. Gene Annotation in Variant Reports

The production panel BED (`All_target_regions_UCL_SingleGeneNIPT_TE-96276661_hg38_*.bed`) is a 4-column Twist file (chrom, start, end, name). The pipeline expects up to 9 columns (chrom, start, end, name, score, strand, gene, disease, pathogenic_variant); missing columns result in empty fields in `variant_report.json` .

FILE	GENE FIELD	HGVS FIELD
<code>variant_report.json</code>	Populated via fallback from BED <code>name</code> col (<code>name.split("_")[0]</code>) — applied as of latest build	Not present
<code>*_final_report.json</code> (post-VEP)	As above	<code>vep.hgvsc</code> , <code>vep.hgvsp</code> (fetal variants only)

